Huawei Access Node JAY-ACC-13 Failure SOP: Addressing ASIC Memory Parity Errors and Line Card Crashes with Power Supply Redundancy Loss

Node: JAY-ACC-13 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2671

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for addressing ASIC memory parity errors, line card crashes, and power supply redundancy loss on the Huawei Access Node JAY-ACC-13 at the Jayanagar Metro Site. This failure domain is rooted in the inherent complexities of the Huawei Access Node's hardware and software architecture, specifically the Line Card's reliance on the ASIC to process forwarding table updates and handle packet switching. When the ASIC experiences memory parity errors or crashes, it can lead to a cascade of events resulting in line card crashes and, ultimately, power supply redundancy loss. The high availability features of the Huawei Access Node, such as the redundant power supply and line cards, are designed to mitigate such failures. However, in the event of a failure, the SOP provides a structured approach to diagnosing and resolving the issue, ensuring minimal downtime and maintaining network availability.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that the remote hands are on standby for assistance.

- Check the redundant link status to ensure that the backup link is operational and can take over in case of a failure.

- Gather all relevant logs and configuration files for analysis.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display power supply, Expected Output: Power supply status: normal, Description: This command is used to verify the power supply status and ensure that the redundant power supply is operational.

Command: display line-card, Expected Output: Line card status: online, Description: This command is used to verify the line card status and ensure that it is online and operational.
```

```
Command: display asic memory, Expected Output: ASIC memory status:
normal, Description: This command is used to verify the ASIC memory
status and ensure that it is normal and not experiencing parity
errors.
```

Resolution Steps

Step 1: Step: Power Cycle the Line Card, Command: power-cycling line-card 1/1/1, Description: This step involves power cycling the line card to reset it and clear any pending errors. This is a non-intrusive step and does not require any configuration changes.

Step 2: Step: Replace the Faulty ASIC, Command: replace asic 1/1/1, Description: This step involves replacing the faulty ASIC with a new one. This requires a hardware replacement and may require additional configuration changes.

Step 3: Step: Verify Redundant Power Supply, Command: verify power supply, Description: This step involves verifying that the redundant power supply is operational and can take over in case of a failure.

Post-Check Verification

```
Command: display power supply, Expected Output: Power supply status:
normal, Description: This command is used to verify the power supply
status and ensure that the redundant power supply is operational.

Command: display line-card, Expected Output: Line card status: online,
Description: This command is used to verify the line card status and
ensure that it is online and operational.

Command: display asic memory, Expected Output: ASIC memory status:
normal, Description: This command is used to verify the ASIC memory
status and ensure that it is normal and not experiencing parity
errors.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: restore config, Description: This step involves restoring the original configuration to ensure that the system is returned to its previous state.

- Step: Power Cycle the System, Command: power-cycling system, Description: This step involves power cycling the system to ensure that all changes are cleared and the system is restarted in its previous state.